# GC3 Portal Complete Implementation Roadmap

End-to-End Breakdown of All Phases (Phase 0 – Phase 5) and Detailed Steps

---

### Phase 0: Monorepo Architecture & Foundation

- **Step 0.1:** Monorepo Scaffolding (pnpm, Next.js 16, Express 5, packages structure).
- **Step 0.2:** Core Type definitions & Zod validation packages config.

### Phase 1: Identity, Auth & Onboarding

- **Step 1.1:** Database Setup & Double Multi-Tenant Isolation Schema (`organization_id`).
- **Step 1.2:** Authentication & RBAC (JWT sessions, scrypt password hashing, permissions).
- **Step 1.3:** Token-based Invites (Invite acceptance, token rotation, resend & revocation).
- **Step 1.4:** Pluggable Notification System & SMTP Email Driver (Nodemailer against Mailpit).
- **Step 1.5:** Permanent Notification Delivery Audit Log (`NotificationDelivery` table).

### Phase 2: Client Portal & Service Requests

- **Step 2.1:** Service Request & Deliverables Schema Data Layer.
- **Step 2.2:** Service Request API Routes & State Transition Workflow Guards.
- **Step 2.3:** Next.js Frontend Integration (Dashboard, Request Creation & Details Drawers).
- **Step 2.4:** Service Category CRUD Management & Request Editing Policy Guards.
- **Step 2.5:** Staff Deliverable Authoring & Private Internal Notes Barrier.
- **Step 2.6:** Deliverable Versioning, Governance Reset on File Edits, & Delivery History (`/deliveries`).

### Phase 3: Delivery Ops & Workflows

- **Step 3.1:** Staff & Pod Manager Internal Operations Console Layout.
- **Step 3.2:** Task Allocation, Priority Queueing & Specialist Workloads.
- **Step 3.3:** Multi-Pod Workspace Routing & Team Collaboration Tools.
- **Step 3.4:** Operational SLA Tracking & Delivery Milestone Metrics.

### Phase 4: Finance & Subscriptions

- **Step 4.1:** Multi-Tenant Subscription Tier & Billing Limits Schema.
- **Step 4.2:** Invoice Generation, Usage Tracking, & Service Credit Management.
- **Step 4.3:** Client Payment Gateway Integration & Transaction Ledger.
- **Step 4.4:** Financial Reporting & Revenue Analytics Dashboard.

### Phase 5: AI Automation & Agent Guardrails

- **Step 5.1:** Autonomous AI Agent Integration for Lead Processing & Task Parsing.
- **Step 5.2:** Human-in-the-Loop Gateways (Strict approval checkpoints before release).
- **Step 5.3:** Automated Document Processing, Deliverable Checks & Summaries.
- **Step 5.4:** End-to-End Autonomous Workflow Audit Logs & Safety Compliance.